Exact two-particle relaxation crossover for the symmetric inclusion process on hypercubes

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Abstract

The symmetric inclusion process describes diffusion with hopping rates that increase with the occupation of the destination. Its slowest relaxation mode can differ from that of a single particle. We determine the full spectral gap for exactly two particles on every finite Boolean hypercube, with a homogeneous positive site parameter and equal positive edge conductances. The gap is the minimum of the inherited one-particle eigenvalue and the first positive eigenvalue of an explicit birth-death chain for the Hamming distance between the particles. Translation characters exhaust the two-particle state space: every nontrivial character is bounded below by the one-particle gap, while all nonradial modes in the translation-invariant sector lie higher. A rank-one resolvent calculation then gives an exact integral formula for the finite-parameter crossover. A slower separation mode occurs precisely below this positive threshold in dimensions four and higher; the four-dimensional threshold is $1/35$. A recurrence determines the threshold in every dimension and proves its inverse-square dimension scaling. The result specifies when two-particle relaxation is controlled by collective separation rather than a linear occupation observable. It concerns a finite graph with exactly two particles and leaves the general many-particle degree-two gap conjecture unresolved.

Keywords: symmetric inclusion process; spectral gap; hypercube; reversible relaxation; two-particle dynamics.

1 Introduction

The symmetric inclusion process is an established attractive interacting particle model. For conservative dynamics, spectral inheritance makes one-particle eigenvalues available in every particle sector, but does not ensure that they govern relaxation. Kim and Sau proved the one-particle gap identity when all site parameters are at least one [1], and subsequently established its failure on suitable geometries outside that regime [2]. They also proved an asymptotic two-particle reduction as diffusivity vanishes along a fixed positive direction. Their proposed exact identity $\lambda_k = \lambda_2$ for all $k \geq 2$, arbitrary finite connected graphs and positive site parameters remains a conjecture. The later work of Caputo, Quattropani and Sau proves degree-two reduction for stochastic exchange dynamics and explicitly retains the Brownian energy process/inclusion-process question [3, Sec. 1.4.6].

Here we calculate the full two-particle gap on a coherent graph family at finite positive diffusivity. For the Boolean hypercube Q_D, we obtain an exact boundary between one-particle and pair-separation relaxation. The contribution is the full-sector comparison and the finite-parameter boundary in every dimension. Qualitative failure of one-particle reduction is already known from [2]; we do not claim that phenomenon as new. Relative coordinates for two

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interacting particles likewise have prior applications, including the exact one-dimensional transition calculation of Carinci, Giardinà and Redig [4]. The present calculation concerns the finite hypercube and its full two-particle spectral gap. It gives no identity at particle number three or higher.

2 Model and exact result

Write $V = (\mathbb{Z}/2\mathbb{Z})^D$, $D \geq 1$, and join x to $x + e_i$ for each coordinate vector e_i. Addition is modulo two. All edges have conductance $c > 0$, and the site parameter is $a > 0$. For fixed particle number k, the state space is $\Xi_k = \{\eta \in \mathbb{N}_0^V : \sum_x \eta_x = k\}$ and the backward generator is

$$(L_k f)(\eta) = c \sum_{x \in V} \sum_{i=1}^{D} \eta_x (a + \eta_{x+e_i}) \big[f(\eta - \delta_x + \delta_{x+e_i}) - f(\eta) \big]. \tag{1}$$

Terms with $\eta_x = 0$ vanish. This directed jump sum has no factor $1/2$. It conserves particle number, has no reservoirs, and is reversible with probability law

$$\mu_{a,k}(\eta) = \frac{k!}{(2^D a)_k} \prod_{x \in V} \frac{(a)_{\eta_x}}{\eta_x!}, \qquad (a)_m = \prod_{j=0}^{m-1} (a + j), \quad (a)_0 = 1. \tag{2}$$

These conventions agree with the conservative generator and measure of [2, Eqs. (2.1)–(2.2)]. Let λ_k be the smallest strictly positive eigenvalue of $-L_k$ in $L^2(\mu_{a,k})$. All parameters except c are dimensionless; c, the eigenvalues and the gaps have inverse-time units. The relaxation time of the slowest nonconstant mode is $1/\lambda_k$.

For $d \in \{0, \dots, D\}$, define

$$\ell_d = 2da + 2\,\mathbf{1}_{\{d=1\}}, \qquad u_d = 2(D - d)a. \tag{3}$$

The negative birth-death generator $T_D(a)$ has action

$$(T_D(a)h)_d = \ell_d (h_d - h_{d-1}) + u_d (h_d - h_{d+1}), \tag{4}$$

with absent endpoint terms omitted. Write $\rho_D(a)$ for its first positive eigenvalue. Conductance has been set to one in this auxiliary matrix, so that the corresponding physical eigenvalue is $c\rho_D(a)$.

Theorem 1. *For every finite integer $D \geq 1$ and every $a, c > 0$, the model (1) satisfies*

$$\lambda_1 = 2ca, \qquad \lambda_2 = c \min\{2a, \rho_D(a)\}. \tag{5}$$

Define

$$a_D = (D - 1)2^{-(D-1)} \int_0^1 (1 + u^2)^{D-1}\, du - 1. \tag{6}$$

For $D \leq 3$, $\lambda_2 = \lambda_1$ at every $a > 0$. For $D \geq 4$, $a_D > 0$ and

$$\lambda_2 < \lambda_1 \iff 0 < a < a_D, \qquad \lambda_2 = \lambda_1 \iff a \geq a_D. \tag{7}$$

The thresholds obey

$$a_{D+1} = \frac{D^2 a_D + 1}{(D - 1)(2D + 1)} \quad (D \geq 2), \qquad a_4 = \frac{1}{35}, \qquad \lim_{D \to \infty} D^2 a_D = 1. \tag{8}$$

The proof first controls every symmetry sector of the full occupation space, then locates the crossing inside the remaining distance chain. This order is necessary: distance lumping alone supplies an upper bound on λ_2, rather than the equality in (5).

3 Full two-particle sector comparison

Set $c = 1$ throughout the proof. A two-particle configuration is an unordered pair $\{x, y\}$, including the stack $x = y$. The same generator is obtained from labelled positions by giving the first particle a jump $x \to v$ at rate $a + \mathbf{1}_{\{y=v\}}$ on an edge, and similarly for the second particle. On symmetric functions the labelled and occupation generators agree. In particular, a stack has total rate $2a$ to each neighboring destination.

3.1 Translation characters

For $q \in V$, let $\chi_q(x) = (-1)^{q\cdot x}$. Translations preserve the generator and reversible measure, so their character spaces are invariant, orthogonal, and exhaustive. A function in character q can be written as

$$F(\{x, y\}) = \chi_q(x)g(x + y). \tag{9}$$

Exchanging x and y requires $g(z) = \chi_q(z)g(z)$. Thus g is supported on $A_q = \{z : \chi_q(z) = 1\}$. Conversely, this support restriction makes (9) well-defined. With $N = 2^D$, the character dimensions are N at $q = 0$ and $N/2$ at each $q \neq 0$, and their sum is

$$N + (N - 1)\frac{N}{2} = \frac{N(N + 1)}{2} = |\Xi_2|. \tag{10}$$

There is therefore no missing two-particle sector.

Substitution of the labelled jump rates gives the negative-generator block on A_q:

$$(B_q g)(z) = \big(2Da + 2\,\mathbf{1}_{\{|z|=1\}}\big)g(z) - 2\sum_{i:q_i=0}\big(a + \mathbf{1}_{\{z=e_i\}}\big)g(z + e_i), \qquad z \in A_q, \tag{11}$$

where $|z|$ is Hamming weight. If $q_i = 1$, the two off-diagonal contributions cancel, and their departure rates remain in the diagonal. The extra rate at $z = e_i$ describes attraction into a stack. At $z = 0$ the diagonal is $2Da$, with no extra attraction term.

Lemma 1. *For $q \neq 0$, every eigenvalue of B_q is at least $2|q|a$. Characters with $|q| = 1$ attain $2a$.*

Proof. Let $m = |q| \geq 1$ and $r = D - m$. The m coordinates where $q_i = 1$ have even parity in A_q and are unchanged by surviving off-diagonal moves. Fix their bit pattern. If all those bits are zero, the remaining component is

$$B_0(Q_r) + 2maI. \tag{12}$$

Here $B_0(Q_r)$ is itself a translation-invariant block of a reversible Markov negative generator, with nonnegative spectrum and a zero eigenvalue. For $r = 0$ it denotes the one-dimensional zero operator. Any other allowed pattern has at least two nonzero bits, so that neither $z = 0$ nor $|z| = 1$ occurs. Its component is $2a\mathcal{L}_{Q_r} + 2maI$, where $\mathcal{L}_{Q_r}$ is the ordinary cube Laplacian. This proves the lower bound. In the all-zero component with $m = 1$, the constant function in the remaining coordinates attains $2a$. □

The attained eigenfunctions include the linear occupation observables

$$\Phi_i(\eta) = \sum_{x\in V}(-1)^{x_i}\eta_x. \tag{13}$$

Their eigenvalue is $2a$. Independently, the one-particle negative generator is $a\mathcal{L}_{Q_D}$; Walsh characters have Laplacian eigenvalues $2|q|$, giving $\lambda_1 = 2a$.

3.2 The translation-invariant sector

The block B_0 is reversible for the unnormalized weights

$$w(0) = \frac{a+1}{a}, \qquad w(z) = 1 \quad (z \neq 0). \tag{14}$$

One can obtain these weights directly from (2): distinct-position configurations have unnormalized mass a^2 and stacks have mass $a(a+1)/2$, while their translation orbits have sizes $N/2$ and N, respectively. Detailed balance on $0 \leftrightarrow e_i$ also gives $w(0)2a = w(e_i)2(a+1)$.

Coordinate permutations preserve w and B_0. Their invariant subspace $\mathcal{R}$ consists of functions of $|z|$. Its weighted orthogonal complement consists of functions with zero sum on every Hamming shell. In particular, for $f \in \mathcal{R}^{\perp}$,

$$f(0) = 0, \qquad \sum_{|z|=1} f(z) = 0, \qquad \sum_{z} f(z) = 0. \tag{15}$$

Self-adjointness makes this complement invariant. Its weighted norm equals its counting norm, and (11) reduces to

$$B_0 f = 2a\mathcal{L}_{Q_D} f + 2\,\mathbf{1}_{\{|z|=1\}} f. \tag{16}$$

At zero, the equality uses the unit-shell sum in (15). The counting-space gap of $\mathcal{L}_{Q_D}$ is two. Consequently,

$$\frac{\langle f, B_0 f\rangle_w}{\langle f, f\rangle_w} = \frac{2a\langle f, \mathcal{L}_{Q_D} f\rangle + 2\sum_{|z|=1} |f(z)|^2}{\sum_z |f(z)|^2} \geq 4a \quad (0 \neq f \in \mathcal{R}^{\perp}). \tag{17}$$

On $\mathcal{R}$, there are d flips decreasing distance and $D-d$ increasing it, with the additional rate two into zero from distance one. The restriction is therefore exactly (4). Its reversible masses may be chosen as

$$\pi_0 = \frac{a+1}{a}, \qquad \pi_d = \binom{D}{d} \quad (1 \leq d \leq D). \tag{18}$$

All neighboring transition rates are positive, so this finite chain is irreducible. Lemma 1, (17), and the attained eigenvalue $2a$ prove the two-particle formula (5) at $c = 1$.

4 Exact location of the crossover

4.1 A radial resolvent at the competing eigenvalue

Let R_D be the ordinary radial cube Laplacian: its diagonal is D and its downward and upward entries are $-d$ and $-(D-d)$. Its eigenvalues are $0, 2, \ldots, 2D$. For example, the shell sums of Walsh characters of each weight give $D+1$ radial eigenfunctions with these distinct eigenvalues. With ε_d the coordinate vectors of $\mathbb{R}^{D+1}$, (3) gives

$$\frac{T_D(a)}{2a} = R_D + a^{-1}\varepsilon_1(\varepsilon_1 - \varepsilon_0)^{\mathsf{T}}. \tag{19}$$

Since $M = R_D - I$ is invertible, the determinant lemma yields

$$\det(T_D(a) - 2aI) = (2a)^{D+1}\det(M)\left(1 + \frac{h_D}{a}\right), \qquad h_D = (\varepsilon_1 - \varepsilon_0)^{\mathsf{T}} M^{-1}\varepsilon_1. \tag{20}$$

Put $v = M^{-1}\varepsilon_1$. The row at distance zero gives $(D-1)v_0 - Dv_1 = 0$, hence $h_D = -v_0/D$. On the full cube, the unit-shell indicator is $\mathrm{Adj}(Q_D)\delta_0 = (DI - \mathcal{L}_{Q_D})\delta_0$. The corresponding radial equation therefore implies

$$v_0 = (D-1)G_D - 1, \qquad G_D = \left[(\mathcal{L}_{Q_D} - I)^{-1}\right]_{00} = 2^{-D}\sum_{j=0}^{D}\frac{\binom{D}{j}}{2j-1}. \tag{21}$$

The last expression follows by normalized Walsh diagonalization; none of its denominators is zero.

Writing the positive-j terms as integrals gives

$$G_D = 2^{-D}\left[-1 + \int_0^1 \frac{(1+u^2)^D - 1}{u^2}\,du\right]. \tag{22}$$

Integration by parts has a vanishing boundary term at $u = 0$ and yields

$$G_D = -1 + D2^{-(D-1)}\int_0^1 (1+u^2)^{D-1}\,du. \tag{23}$$

Combining (21)–(23) proves $h_D = -a_D$. Thus a radial eigenvalue can equal $2a$ at a positive parameter only when $a = a_D$. If $a_D > 0$, the zero of (20) as a function of a is simple.

4.2 Which eigenvalue crosses, and in which direction?

For every $a > 0$, $T_D(a)$ is similar by (18) to a real symmetric tridiagonal matrix with nonzero neighboring entries. Its eigenvalues are real, simple and continuous in a, and exactly one is zero. Moreover, $T_D(a)/a \to 2R_D$ as $a \to \infty$. Every positive radial eigenvalue is therefore larger than $2a$ for sufficiently large a.

If $a_D \le 0$, the determinant has no positive-parameter zero, so none of these eigenvalues can cross $2a$; in particular $\rho_D(a) > 2a$ for all $a > 0$. Suppose instead $a_D > 0$. A higher positive radial eigenvalue cannot be the one equal to $2a$ at a_D: the first positive one would then be strictly below $2a$ there, forcing another crossing between a_D and the large-a regime. This contradicts the unique possible zero of (20). Hence the crossing belongs to ρ_D. The simple determinant zero, together with eigenvalue simplicity, enforces a sign change. Continuity and the absence of any other crossing show that

$$\rho_D(a)\begin{cases} < 2a, & 0 < a < a_D, \\ = 2a, & a = a_D, \\ > 2a, & a > a_D. \end{cases} \tag{24}$$

No monotonicity assumption on $\rho_D(a)/a$ is needed.

4.3 Threshold signs, recurrence and dimension scaling

Define $H_m = 2^{-m}\int_0^1 (1+u^2)^m\,du$. Integrating the derivative of $u(1+u^2)^m$ gives

$$H_m = \frac{1 + mH_{m-1}}{2m+1}, \qquad H_0 = 1. \tag{25}$$

Since $a_D = (D-1)H_{D-1} - 1$, this identity implies (8). Direct evaluation gives

$$a_1 = -1, \quad a_2 = -\frac{1}{3}, \quad a_3 = -\frac{1}{15}, \quad a_4 = \frac{1}{35}, \quad a_5 = \frac{17}{315}, \quad a_6 = \frac{37}{693}. \tag{26}$$

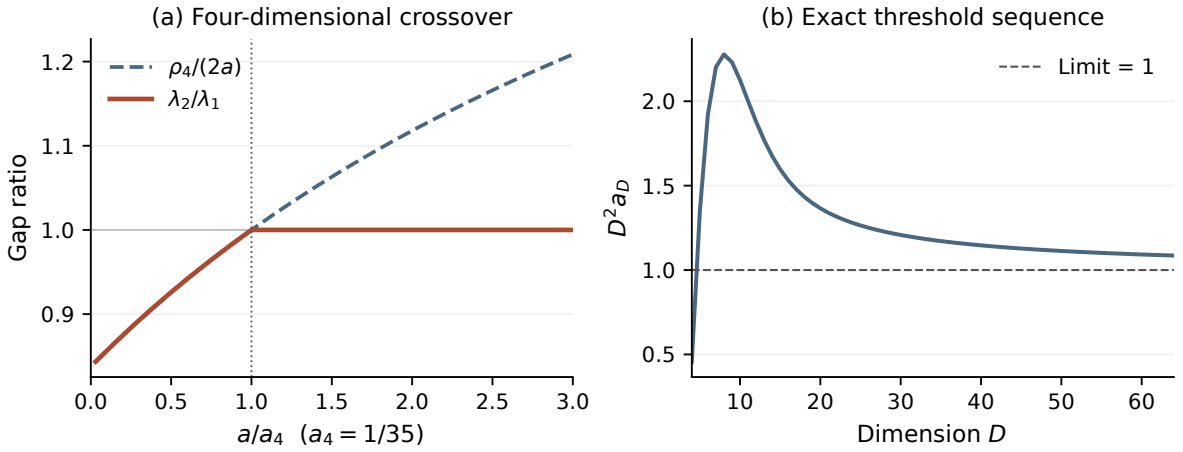


Figure 1: The exact two-particle reduction. Left: the radial gap divided by the one-particle gap on Q_4, and the full two-particle ratio obtained by taking its minimum with one. The vertical line is the exact crossing $a/a_4 = 1$, with $a_4 = 1/35$. Eigenvalues are evaluated at 80-decimal working precision for illustration; the crossing is established analytically. Right: $D^2 a_D$ from the rational recurrence, for $4 \le D \le 64$, with its proved limiting value one. Neither panel asserts a thermodynamic phase transition.

The recurrence propagates positivity from $D = 4$. Together with (24) and (5), this proves the crossover statements.

For the asymptotic, put $b_D = D^2 a_D$. Then

$$b_{D+1} = q_D(b_D + 1), \qquad q_D = \frac{(D+1)^2}{(D-1)(2D+1)} \longrightarrow \frac{1}{2}. \tag{27}$$

For $D \ge 6$, $0 < q_D \le 4/5$, so the recurrence first bounds b_D uniformly. Also,

$$|b_{D+1} - 1| \le \frac{4}{5}|b_D - 1| + |2q_D - 1|. \tag{28}$$

Iteration gives a geometrically decaying initial contribution and a convolution of a summable geometric sequence with a term tending to zero. Both vanish as $D \to \infty$, proving $b_D \to 1$. Finally, restoring c multiplies every eigenvalue by c and leaves a_D unchanged. This completes the proof of Theorem 1.

5 Four-dimensional example and reproducibility

For $D = 4$, the radial characteristic polynomial, with the convention $p_4(\lambda) = \det(\lambda I - T_4(a))$, is

$$p_4(\lambda) = \lambda(\lambda - 8a)\big[\lambda^3 - (32a + 2)\lambda^2 + (304a^2 + 32a)\lambda \\ - 768a^3 - 48a^2\big]. \tag{29}$$

In particular $p_4(2a) = 96a^4(35a - 1)$. Below $a_4 = 1/35$, $\rho_4(a)$ is the smallest positive root of the cubic factor, and $\lambda_2 = c\rho_4(a) < 2ca$. Figure 1 illustrates this crossing and the dimension scaling of the exact threshold.

The accompanying symbolic verification compares full occupation generators assembled from direct jump rates with a separate reversible incidence construction. It checks character embeddings, radial intertwiners and exact root ordering at $D = 3$, $a = 1/100$, and at $D = 4$,

$a = 1/100, 1/35, 1/10, 1$. Resolvent, integral and recurrence identities are also compared exactly for $1 \le D \le 16$. These finite checks support the calculation; the analytic sector bounds and crossing argument establish the unrestricted finite-D statement. The figure-generation script writes the plotted data and compares high-precision radial eigenvalues with a separately assembled double-precision symmetric matrix. Those numerical values are illustrations, not exact certificates. Regeneration instructions and package versions are supplied with the reproducibility material.

6 Relaxation modes and scope

Below a_D, a translation-invariant function of particle separation is the slowest two-particle mode. The bounds on the other sectors ensure that its slower relaxation is not an artifact of radial projection. At and above the threshold, a linear occupation observable such as (13) attains the gap. The result therefore distinguishes pair-separation relaxation from a mode already visible to a single particle, and locates the distinction at finite diffusivity.

Published spectral monotonicity gives $\lambda_k \le \lambda_2$ for every $k \ge 2$ [2, Eq. (2.7)]. Hence below the threshold one-particle reduction is excluded in all those sectors. This observation uses an existing monotonicity result and gives no equality between λ_k and λ_2. Above the threshold, the many-particle gap may still be smaller than the one-particle gap. The general degree-two conjecture is thus left unresolved.

All conclusions assume exactly two particles for the calculated gap, strictly positive homogeneous site parameter, equal positive edge conductances, and a finite closed cube. Heterogeneous weights, other geometries and reservoir dynamics are outside the result. The limit $D^2 a_D \to 1$ describes the sequence of two-particle crossover boundaries; it is not a statement about a thermodynamic transition or a uniform many-particle limit.

Data, code and use of generative AI

Reproducibility scripts, exact verification outputs, plotted data and environment requirements accompany this manuscript. No experimental dataset is used. OpenAI Codex was used in the research process for candidate exploration, analytic derivations, code generation, symbolic and numerical checking, literature comparison, and manuscript drafting. Separate AI-assisted reconstruction and checks were performed; they do not constitute external peer review. AI tools are not authors. Human scientific review and approval of this draft remain pending.

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